

Owen Karmel

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Education

Cornell University, Ithaca NY

Aug 2023 - May 2027

- Pursuing B.A. in physics with dual minors in biology and machine learning.
- Current GPA: 3.85

Research Experience

Research Assistant, Bioengineering, Ag Robotics Lab, Cornell University

Sep 2026 - Current

- Machine vision for autonomous navigation between orchard rows.

Research Assistant, Biomedical Engineering, Jiang Lab, Cornell University

Aug 2024 - Current

- Submitting a first-author manuscript in controllable self-amplifying RNA to the journal ACS Synthetic Biology.
- Led two independent projects to (1) construct miRNA sponges targeting miRNAs implicated in senescence, and (2) design a novel self-amplifying mRNA platform.
 - Designed and tested dozens of custom DNA plasmids, including eight unique miRNA-responsive fluorescent reporters.
 - Investigated the integration of two riboswitches into self-amplifying RNA via flow cytometry to enable prolonged, non-immunogenic expression.
 - Optimized low-cost stem-loop-mediated RT-qPCR in a series of 10+ experiments to accurately measure senescence-associated miRNAs across three cell types.
- Developing mRNA-based therapeutic approaches that selectively target senescent cells as a novel strategy to address aging and age-related diseases.
- Supported a diagnostic project utilizing Raman spectroscopy and machine learning to identify senescent cells and detect metabolic changes in a rapid, label-free manner.

Research Intern, Thymune Therapeutics

Jun 2026 - Current

- Optimized generation of thymus cells and developed assays for their functional characterization.
- Built internal website for scRNA-seq data visualization with Azure DevOps and bash scripting.

Engineering Intern, Regeneron Pharmaceuticals

May 2024 - Aug 2024

- Tested 160+ device samples in a 3-month accelerated aging study of pre-filled syringe shelf life.
- Discovered via ANOVA a significant increase in plunger depression force and proposed a mechanistic explanation; presented results to department leadership in a 15-page technical report.

Research Assistant, Biomedical Engineering, Cosgrove Lab, Cornell University

Dec 2023 - May 2024

- Assisted in FACS to isolate primary macrophages by identifying markers from scRNA-seq.

Honors & Awards

Cornell Rawlings Presidential Research Fellowship

2025

- Supports top 2% of Cornell undergraduates who have a strong commitment to research with a \$7,000 research and conference travel support account.

Jiang Fellowship

2025

- Highly selective \$11,000 award supporting Cornell students to perform summer research at a startup company of their choosing.

Cornell Summer Experience Grant

2025

- Competitive \$5,000 award for pursuing summer research.

Cornell Einhorn Center Engaged Student Grant

2024

- \$1,000 award for commitment to advocacy/outreach, e.g. longevity research advocacy.

Service & Leadership

Volunteer, Longevity Biotechnology Association

Nov 2024 - Current

- Wrote a report evaluating 44 relevant companies for membership to promote outreach and investment in the longevity field.
- Photography and social media lead for the 2025 LBA Investors Conference in NYC.

Teacher, Splash! at Cornell

Dec 2025 - Current

- Taught hands-on classes in molecular biology and rocketry to middle and high schoolers.

Founder and co-President, Cornell Longevity Association

Dec 2025 - Current

- Organized two speaker events with biotech CEOs and a director of geriatric medicine, each with 20-60+ attendees; built a custom website for the Association.

Editor, Cornell Undergraduate Research Journal

Aug 2025 - Aug 2026

- Peer-reviewed undergraduate research paper submissions in all disciplines.

Peer Mentor, Cornell University

Oct 2024 - Current

- Served as a mentor for 6 undergraduates in professional development during the CURB Peer Mentorship Program.
- Led 14-mile summit of Algonquin and Cascade mountains in Adirondacks in a NOLS expedition.

Presentations

Karmel, O. et al. (2026, October 24). *Tetracycline-responsive aptazymes enable tunable expression of self-amplifying RNA* [Presentation]. Biomedical Engineering Society Annual Meeting, Orlando, FL, United States.

Karmel, O., & Beaman, H. (2026, August 12). *scRNA-seq visualization website* [Presentation]. Thymune Company-Wide Meeting, Boston, MA, United States.

Karmel, O. (2024, August 7). *Characterization of a novel accelerated aging strategy for assessing device functional stability* [Presentation and technical report]. Combination Product Dept. Leadership at Regeneron, Albany, NY, United States.

Projects

OpenFront Reinforcement Learning Agent

- Training a deep RL agent to play OpenFrontIO (an online real-time strategy game) that can beat the "medium" difficulty Nation bot.

Proxmox Homelab

- Configured GPU/resource passthrough, containerization, router VPN, and local network.
- Deployed a remote development environment and containerized instances of OpenWebUI and Ollama (local LLM server) for personal use.

Additional Skills

Wet Lab Skills: Mammalian cell culture (primary human dermal fibroblasts, NIH3T3s, HEK293s), restriction enzyme and Gibson cloning, flow cytometry (gene expression, SA- β -gal assay, intracellular and extracellular antibody panels), in vitro transcription, RT-qPCR (for both mRNA and miRNA), western blotting, nanoparticle tracking analysis.

Computational Skills: Proxmox (Debian, LXC containerization), Bash, Azure DevOps and Static Web Apps, and Python (NumPy, Pandas, scikit-learn, Torch), Affinity Designer and Graphpad Prism for scientific graphics, R (Seurat), Benchling, JMP for statistical analysis.

Coursework: Organic Chemistry, Stem Cell Biology, Machine Learning, Reinforcement Learning, Computational Genetics and Genomics, Anatomy and Physiology, Advanced Electromagnetism.